

Exact Memory-Time Optimization for Prefix-Cached Language Model Serving

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Abstract

Retaining language-model prefix states trades recomputation against storage time. Optimizing each cached block independently can overcount savings: a resident block is usable only when the required preceding prefix is also available. We introduce Prefix-Certificate Retention (PCR), an exact finite-trace formulation for static, grouped, reset-on-access timeouts. Usable-prefix rewards become nodes whose prerequisites are timeout thresholds and preceding hit certificates. The resulting maximum-weight closure reduces to one minimum cut, with graph size linear in the number of block lookups and timeout choices. A breakpoint theorem extends the construction to all nonnegative timeouts without discretization error. We also derive a linear-time-in-grid-size dynamic program for ordered timeouts and bounds that certify the cost of this restriction. Exhaustive small-instance checks and chronological replay of 39,632 public Mooncake requests validate the formulation. On the fixed grid, ordered timeouts attain the unrestricted training optimum in 118 of 120 trace-grouping-price cases. Heterogeneous retention improves several held-out memory-time tradeoffs, but finer training optimization does not uniformly improve transfer. The contribution is a tractable optimization model and an auditable benchmark for retention policies; the experiments measure usable prefix blocks and storage time, not GPU latency.

Keywords: prefix caching; maximum closure; time-to-live optimization; language model serving; memory-time budgets.

Code and reproducibility:

github.com/shi1720/prefix-certificate-retention

1 Introduction

Language-model services repeatedly process shared system prompts, conversation histories and agent instructions. Retaining their key-value (KV) states can avoid repeated prefill work, but every retained block occupies storage until it is reused or discarded. PagedAttention, RadixAttention and disaggregated cache architectures provide practical mechanisms for this reuse [6, 10, 12]. They also create an operational decision: how long should different parts of a prefix remain available?

The optimization is subtle even before queueing and hardware constraints enter. Consider a request for blocks (A, B) . If B remains in an independently addressed store but A has expired, counting B as a saved prefix block is incorrect for a serving interface that resumes only from a contiguous cached prefix. Requiring the timeout of A to be at least that of B prevents this inconsistency. However, that requirement can retain A unnecessarily: other requests may refresh A while B waits for reuse. Thus, independent hit accounting can be optimistic, whereas universally prefix-preserving timeout ordering can be conservative.

This paper asks three questions. Can actual prefix reuse be optimized exactly without imposing timeout order? Can the gap between ordered and unrestricted retention be certified?

And do the resulting profiles improve memory-time tradeoffs on later requests? The first two questions admit deterministic answers within a precise model. The third requires empirical evaluation and has a mixed answer.

We study static group timeouts fitted to a historical trace. The objective is the number of reusable full prefix blocks minus a declared storage price times occupied block-seconds. It is a configuration objective for a cache store, not a complete inference-system objective. The main contributions are:

1. An exact representation of contiguous-prefix reuse by a directed acyclic graph of hit certificates. Combining this representation with incremental storage costs yields a maximum-weight closure problem, solved by a standard minimum-cut algorithm (Theorem 2).
2. A reuse-age breakpoint theorem that removes the timeout-grid restriction while retaining a graph of linear size (Theorem 3).
3. A coherent-policy dynamic program and a three-way optimality bound, together with a strict counterexample and expected-budget certificates (Section 5).
4. Reproducible tests and public-trace experiments separating mathematical optimality, the practical value of a simpler restriction, and held-out transfer (Sections 6–7).

Maximum closure and its reduction to minimum cut are classical [9]. Our contribution is the prefix-retention formulation, its exact accounting and breakpoint structure, and the resulting certification study. Neither differentiated timeouts nor prefix-aware cache management is claimed as new.

2 Relation to prior work

Timer-based cache optimization. Ferragut, Rodriguez and Paganini characterize timer policies through arrival distributions and hazard rates [2]. Soft-TTL optimizes time-varying fractions of retained files [4]. These results establish substantial precedent for retention optimization. Here the input is a finite trace, the policy is a static timeout per block group, and the reward includes conjunctions of prefix availability. We do not require a renewal model or estimate hazard rates.

Prefix and agent-serving systems. PagedAttention addresses KV memory management [6]; SGLang’s RadixAttention exploits shared prefixes [12]. CachedAttention retains conversation state across requests [3], while Mooncake integrates a shared KV store with disaggregated serving [10]. Recent agent-serving work includes workflow-level scheduling in SAGA [5] and execution-transition-guided eviction and prefetching in CacheScout [11]. PCR supplies an off-line optimization benchmark for one retention subproblem. It does not replace their scheduling, transfer or execution models.

The closest configuration formulation. Kareto groups frequent prefix subtrees, models timeout-dependent block hits and storage exposure, and uses multi-start sequential quadratic programming for a budgeted group-timeout allocation [13]. Grouping by popular prefixes and fitting timeouts from historical accesses are therefore explicit antecedents. PCR differs by representing usable-prefix conjunctions directly, censoring occupancy at the observation horizon, and solving a priced static-timeout problem globally. Our group baseline is a controlled simplification inspired by this literature, not a reproduction of Kareto’s tiered-storage system or its reported latency results.

Dependency constraints and graph optimization. Dependency-aware online caching studies cache decisions with object prerequisites and hard-capacity constraints [1]. Our un-

restricted model permits orphaned stored blocks but awards reuse only to contiguous prefixes; its constraint is priced memory-time. This difference is what allows temporal refreshes to make unordered timeouts useful. Picard’s maximum-closure reduction [9] provides the algorithmic foundation once the correct hit prerequisites have been identified. The distinction is in the cache model, not in a new general-purpose flow algorithm. A dated literature matrix and search scope accompany the code.

3 Trace model and exact accounting

Let the observation window be $[0, T]$. A request r arrives at time t_r and contains an ordered list of full blocks $(b_{r,1}, \dots, b_{r,L_r})$. A block ID identifies the entire prefix ending at that block; each non-root block has a unique predecessor. The total number of block lookups is $M = \sum_r L_r$. Blocks are mapped to G fixed groups by $g(b)$. One timeout $\tau_g \geq 0$ is used for every block in group g .

The store starts empty. All requests at a common timestamp read the pre-batch state before any of them writes. After those reads, every full block in the batch is available and its expiration is reset to $t + \tau_{g(b)}$. Zero timeout means no retention. A positive timeout remains valid at its expiration instant. These conventions make tied arrivals and exact boundary hits unambiguous. Physical storage is keyed by cumulative hashes and may contain a block whose predecessor is absent. The serving interface counts only the longest contiguous available prefix; it does not reuse an isolated suffix after recomputing a missing ancestor.

Let $a_{r,d}$ be the time since the most recent *strictly earlier timestamp batch* containing $b_{r,d}$, with $a_{r,d} = \infty$ on its first access. Then local availability and usable-prefix availability are

$$x_{r,d}(\tau) = \mathbf{1}\{0 < a_{r,d} \leq \tau_{g(b_{r,d})}\}, \quad (1)$$

$$z_{r,d}(\tau) = \prod_{j=1}^d x_{r,j}(\tau), \quad H(\tau) = \sum_r \sum_{d=1}^{L_r} z_{r,d}(\tau). \quad (2)$$

The strict inequality in (1) follows from batch semantics; it does not discard reuse across distinct positive-age batches. More generally, each $z_{r,d}$ can have an exogenous nonnegative reward $w_{r,d}$. All reported experiments use $w_{r,d} = 1$.

For a physical block b with distinct access timestamps $s_{b,1} < \dots < s_{b,m_b}$, set $s_{b,m_b+1} = T$. Its occupied time is

$$C_b(\tau) = \sum_{j=1}^{m_b} \min\{\tau, s_{b,j+1} - s_{b,j}\}. \quad (3)$$

Define $C_g(\tau) = \sum_{b:g(b)=g} C_b(\tau)$ and $C(\boldsymbol{\tau}) = \sum_g C_g(\tau_g)$. The priced objective is

$$\max_{\boldsymbol{\tau} \in \mathcal{T}} U_\lambda(\boldsymbol{\tau}) := H(\boldsymbol{\tau}) - \lambda C(\boldsymbol{\tau}), \quad \lambda \geq 0. \quad (4)$$

The price has units of saved blocks per occupied block-second. Mean occupancy is C/T . For a fixed model whose full blocks occupy q bytes, byte-seconds are qC ; converting rewards to money or latency requires additional measurements.

Lemma 1 (Exact occupied time). *Equation (3) equals the integral of the block’s residency indicator over $[0, T]$, including when accesses have multiplicity within a batch.*

Proof. Partition the window after the first access into $[s_{b,j}, s_{b,j+1})$. On each interval there is no intervening reset. The block is resident for exactly the smaller of its timeout and the interval length. Summing these disjoint contributions gives (3). A batch writes one physical copy regardless of how many requests read it. The final interval ends at T , so no storage outside the observation window is charged. $\square$

The assumptions are consequential. New blocks are available immediately after their arrival batch, so this is a zero-service-time cache replay. A real system must wait for computation and transfer. Cache pinning by active requests, decode state, admission control, model-version invalidation and capacity-triggered eviction are outside (4). Changing those semantics can invalidate both the additive cost and the threshold representation.

4 Prefix certificates and the minimum-cut formulation

For group g , let its finite ordered timeout choices be $0 = q_{g,0} < q_{g,1} < \dots < q_{g,K_g}$. A shared grid is a special case. Introduce threshold variables $y_{g,k} = \mathbf{1}\{\tau_g \geq q_{g,k}\}$ for $k \geq 1$, with prerequisites $y_{g,k} \Rightarrow y_{g,k-1}$ for $k > 1$. The incremental costs

$$\Delta C_{g,k} = C_g(q_{g,k}) - C_g(q_{g,k-1}) \geq 0 \quad (5)$$

give $C_g(\tau_g) = \sum_{k=1}^{K_g} \Delta C_{g,k} y_{g,k}$.

For a finite age $a_{r,d}$, define $\kappa(r,d) = \min\{k \geq 1 : q_{g(b_{r,d}),k} \geq a_{r,d}\}$. If this set is empty, or any earlier block in the request has no feasible local hit, that depth and all later depths cannot contribute and are omitted. Otherwise introduce a hit node $z_{r,d}$ with implications

$$z_{r,d} \Rightarrow y_{g(b_{r,d}),\kappa(r,d)}, \quad z_{r,d} \Rightarrow z_{r,d-1} \quad (d > 1). \quad (6)$$

Give hit nodes weight $w_{r,d} \geq 0$ and threshold nodes weight $-\lambda \Delta C_{g,k}$. A *closed* node set contains every prerequisite of every node it contains. Choosing a maximum-weight closed set exactly balances the rewards against the timeouts required to make them usable.

Theorem 2 (Exact grid optimization). *Under the model of Section 3, maximizing (4) over the finite product grid is a maximum-weight closure problem with $O(M + \sum_g K_g)$ vertices and arcs. It is solved by one minimum s - t cut. For unit rewards, its optimal value equals $W - F$, where W is the sum of all hit-node weights and F is the minimum-cut value.*

Proof. Any timeout profile selects an initial segment of each threshold chain. Adding exactly the usable hit nodes yields a closed set: the local threshold and all earlier hit certificates exist whenever a prefix is usable. The node-weight sum is its utility by telescoping the incremental costs.

Conversely, a closed set selects initial threshold segments and thus a valid timeout profile. It cannot contain an unusable hit node, because recursively following (6) enforces every required local threshold. Adding any omitted usable hit node and its hit prerequisites does not require new thresholds and cannot reduce weight. Hence a maximum-weight closed set attains exactly the best timeout utility.

For completeness, add an arc from the source to every positive-weight node with capacity equal to its weight, and an arc from every negative-weight node to the sink with capacity equal to the negative of its weight. Every prerequisite arc gets capacity $Q > W$. The all-sink cut has value W , so a minimum cut cannot sever a prerequisite arc. Its source side is therefore closed. Its cut capacity is W minus the source-side node weight, proving the identity. There is one threshold node per positive choice, at most one hit node per lookup, and at most two prerequisite arcs per hit node. This is the classical closure reduction [9] applied to the constructed cache graph. $\square$

The graph-size claim is not a linear-time bound for maximum flow. Building threshold indices by binary search takes $O(M \log(1 + \max_g K_g))$ time once the access ages and grids are available. Identical certificates can be merged: a certificate is keyed by its predecessor certificate, group, and threshold index. Summing their rewards preserves every profile's utility. Our implementation uses this compression.

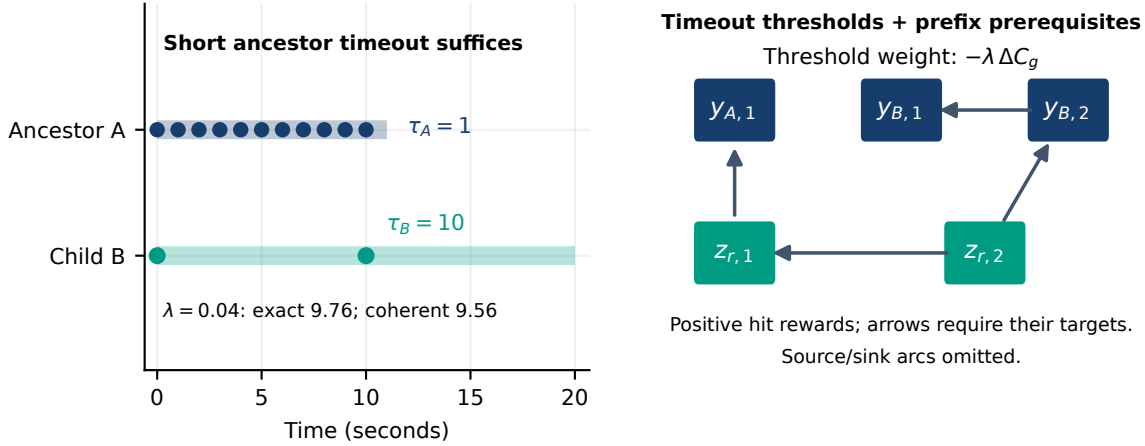


Figure 1: Why prefix certificates are needed. Left: requests for (A, B) at times 0 and 10, and for A at times 1 through 9. A short timeout for A can coexist with a longer timeout for B because A is refreshed. Right: each reward requires its local timeout threshold and the preceding prefix reward. Arrows denote prerequisites. The graph schematic is illustrative; the strict numerical example is evaluated at horizon 20.

4.1 Removing the discretization restriction

Let $\mathcal{A}_g$ contain all finite positive reuse ages of blocks in group g on the trace, even when the corresponding full prefix cannot be reused. Set $\mathcal{Q}_g = \{0\} \cup \mathcal{A}_g$.

Theorem 3 (Reuse-age sufficiency). *For every $\tau \in \mathbb{R}_+^G$, there exists $\tau' \in \prod_g \mathcal{Q}_g$ with $H(\tau') = H(\tau)$ and $C(\tau') \leq C(\tau)$. Consequently, for every $\lambda \geq 0$, the continuous nonnegative-timeout problem has an optimum on these finite grids. Its closure graph has $O(M + G)$ vertices and arcs.*

Proof. For each group, round its timeout down to the largest member of $\mathcal{Q}_g$ not exceeding it. Every reuse-age comparison has the same truth value before and after rounding: if an age was at most the old timeout, that age itself is an eligible grid member and is no larger than the rounded timeout. False comparisons remain false. All local indicators, and hence their prefix products, are unchanged. Each C_g is nondecreasing, so cost cannot increase. There are at most M finite reuse ages before deduplication. The graph bound follows from Theorem 2. $\square$

Cost tables on these grids need not be formed by a quadratic scan. For a group, collect the exposure lengths in (3), sort them as $e_1 \leq \dots \leq e_n$, and store prefix sums $S_j = \sum_{i \leq j} e_i$. If $j(q)$ is the number of exposures strictly below q , then

$$C_g(q) = S_{j(q)} + (n - j(q))q. \quad (7)$$

Sorting ages and exposures gives $O(M \log M + G)$ preprocessing and linear graph storage. The experiment implementation uses integer microseconds for threshold comparisons and evaluates (7) in block-seconds.

5 Coherent policies, bounds and budgets

Suppose the group relations form a forest. If a physical block's predecessor is in a different group, its group is the parent of the block's group; groups may also contain consecutive blocks. Write $p(g)$ for a group's parent. A *coherent* profile satisfies $\tau_g \leq \tau_{p(g)}$ whenever a parent exists.

Proposition 4 (A sufficient invariant, not a necessary trace condition). *Coherent timeouts ensure that every resident block has its whole preceding prefix resident, for every trace satisfying the model. Without this ordering, the invariant need not hold. However, ordering can be strictly suboptimal for the finite-trace utility.*

Proof. Every child access is also an access to its predecessor. At any time, the most recent predecessor reset is at least as recent as the child’s. Its timeout is also at least as long, so child residency implies predecessor residency; induction gives the full prefix. If a child’s timeout exceeds its predecessor’s, one access followed by silence produces an interval with an orphaned child.

For strict suboptimality, use the trace in Figure 1, horizon $T = 20$, choices $\{0, 1, 10\}$, and $\lambda = 0.04$. The unrestricted profile $(\tau_A, \tau_B) = (1, 10)$ gives $H = 11$, $C = 31$ and utility 9.76. Among the six ordered profiles, the best is $(1, 0)$, with $H = 10$, $C = 11$ and utility 9.56. The profile $(10, 10)$ has utility $11 - 0.04(40) = 9.40$. Exhaustion of the remaining ordered profiles proves the strict gap. $\square$

Let $h_g(k)$ count local block hits for group g at common-grid label k , with request multiplicity retained, and let $c_g(k) = C_g(q_k)$. Under coherence these local hits equal the usable hits. For a group v , define

$$D_v(k) = h_v(k) - \lambda c_v(k) + \sum_{u:p(u)=v} \max_{0 \leq j \leq k} D_u(j). \quad (8)$$

The optimum is the sum of $\max_k D_v(k)$ over roots. A postorder traversal with prefix maxima and traceback takes $O(G(K+1))$ time and storage. This follows by conditioning on the parent’s label and optimizing independent child subtrees. Smallest maximizing labels resolve ties.

Define U_{coh}^* as the coherent optimum, U_{PCR}^* as the unrestricted usable-prefix optimum, and the independently optimized local-hit score

$$U_{\text{ind}}^* = \sum_g \max_k \{h_g(k) - \lambda c_g(k)\}. \quad (9)$$

All three quantities in the following bound use the same finite grid and group assignment.

Corollary 5 (Certification of a simpler policy).

$$U_{\text{coh}}^* \leq U_{\text{PCR}}^* \leq U_{\text{ind}}^*. \quad (10)$$

Thus $U_{\text{ind}}^* - U_{\text{coh}}^*$ is a computable upper bound on the price of coherence. If it vanishes, the coherent policy is unrestricted-optimal.

Proof. The coherent feasible set is a subset of the unrestricted set. On any profile, $\sum_{r,d} z_{r,d} \leq \sum_{r,d} x_{r,d}$, while costs are identical. Maximizing the latter separable objective proves the upper bound. $\square$

The raw-hit bound need not be close. If a root is never retained but many deeper blocks survive until a second request, raw local hits can be positive while usable-prefix hits are zero. This is an accounting counterexample, not an empirical claim that the released traces exhibit arbitrary distortion.

5.1 Expected memory-time budgets

For any deterministic policy with $C \leq B$ and any price $\lambda \geq 0$,

$$H \leq U_{\text{PCR}}^*(\lambda) + \lambda B. \quad (11)$$

This is a Lagrangian upper bound. It does not imply that a single priced policy solves every deterministic budget constraint.

Trace	Requests	Full lookups	Unique blocks	Duration (s)	Train/validation/test requests
Conversation	12,031	276,491	170,899	3,537.0	4,401 / 2,447 / 5,183
Tool agent	23,608	386,041	159,983	3,537.0	8,779 / 4,832 / 9,997
Synthetic	3,993	117,888	40,148	1,022.0	1,512 / 807 / 1,674

Table 1: Public traces after discarding incomplete final blocks. Splits are by elapsed time, not request count. There are 39,632 requests in total, of which 35,639 are production-derived.

Proposition 6 (Optimal mixtures for an expected budget). *Fix a finite policy class containing no retention. Permit a policy to be drawn once before replay, independently of the trace, and require $\mathbb{E}C \leq B$. The optimal expected reward is the nondecreasing upper boundary of the convex hull of achievable (C, H) pairs. An optimum uses at most two policies. On a supported segment, (11) is tight with the corresponding supporting price.*

Proof. A distribution on policies has expected (C, H) equal to a convex combination of their pairs; every such combination is implementable by a draw before replay. Maximizing expected reward is a linear program with one normalization equality and one budget inequality. An extreme optimal solution has at most two positive probabilities (one if the budget is slack). A supporting line of the efficient upper hull has nonnegative slope and yields the stated price certificate. $\square$

For adjacent candidate points (C_a, H_a) and (C_b, H_b) , query the optimizer at $\lambda = (H_b - H_a)/(C_b - C_a)$. If its utility exceeds their supporting line, insert the returned point and recurse; otherwise the segment is certified. Our released frontier implementation applies this procedure to the coherent dynamic program. The same argument allows the closure solver as its oracle. Randomizing whole profiles across independent windows is different from switching timeouts midway through a live window, which would change the cache state. Neither the mixture nor a mean-occupancy target guarantees an instantaneous memory limit.

6 Computational study

6.1 Data, endpoint and chronology

We use the public Mooncake FAST’25 traces [7, 10]: two production-derived workloads, `conversation` and `toolagent`, and one synthetic workload. Table 1 gives their sizes. The release is pinned by Git revision and SHA-256 digests. Every hash path is checked for the unique-predecessor property. Hashes represent 512-token blocks; a final partial block is removed because it cannot be counted as a reusable full block.

Each trace is split chronologically into 40% training, 20% validation and 40% test by its last arrival time. Each replay starts empty and stops at its own window boundary. Timeouts and group keys are fitted on training only. The primary endpoint is $(H - \lambda C)/M$ on test; secondary measurements are H/M , mean resident blocks C/T , peak blocks and fitting time. A paired resampling of 60-second test buckets gives descriptive 95% intervals for fixed-policy utility differences (10,000 resamples, seed 20260923). There are only 24 buckets on each production-derived test trace and seven on the synthetic trace. These intervals are neither deployment-level replications nor simultaneous confidence bands across prices, and may understate uncertainty under longer-range dependence.

A local analysis plan was committed before the first policy comparison. It specified the data, time split, grids, primary coherent-policy selection and baselines. After inspecting that study, the certificate formulation, timestamp coalescing and continuous-timeout analyses were added

as explicitly exploratory extensions. They use the same held-out segments, so the paper does not describe them as a fresh confirmatory evaluation or an external preregistration.

6.2 Policy classes and baselines

The fixed timeout grid, in seconds, is

$$\{0, 0.001, 0.1, 0.3, 1, 3, 10, 30, 60, 120, 300, 600, 900\}.$$

All policies are evaluated at every price in

$$\{0, 10^{-4}, 3 \cdot 10^{-4}, 10^{-3}, 3 \cdot 10^{-3}, 10^{-2}, 3 \cdot 10^{-2}, 10^{-1}, 0.3, 1\}.$$

No single price is chosen as the primary endpoint after seeing the results. *Global TTL* fits one timeout. *Group TTL* shares one timeout across first blocks and assigns a timeout to each of the eight most frequent second-block prefixes and a pooled fallback; child timeouts are bounded by the root timeout. *Depth TTL* uses one chain with depth bands

$$[1], [2], [3, 4], [5, 8], [9, 16], [17, 32], [33, 64], [65, 128], [129, \infty).$$

Group-depth TTL uses the shared root and one eight-band chain per prefix group, totaling 73 labels when all eight selected prefixes exist. In every case the root label can cover many distinct physical root blocks. Unseen second-block hashes use the training-defined fallback.

The primary *selected coherent* policy chooses depth or group-depth using validation utility at each price; ties favor depth. All profiles remain trained on training only. PCR is evaluated for each of the four groupings without the coherence restriction. An independent-local-hit optimizer is retained as an accounting diagnostic, with actual prefix hits replayed afterward. The continuous-timeout extension evaluates group-depth PCR at all observed training reuse ages. It also allows ages above 900 seconds, so its comparison with the fixed grid combines finer resolution and a larger maximum timeout.

Capacity-constrained least-recently-used (LRU) and first-in-first-out (FIFO) baselines use 256, 1,024, 4,096 and 16,384 blocks. They have the same batch read semantics and usable-prefix hit metric, with stable trace order for post-batch insertions. These are hash-store reference policies, not replicas of a particular serving engine’s radix eviction. Their hard capacity constraint differs from priced memory-time. A no-eviction replay provides the cold-window reuse reference.

6.3 Verification and numerical implementation

The 11-test suite includes the following independent checks:

- 140 random forests with up to seven nodes and four labels: exhaustive feasible-profile enumeration versus the dynamic program and a separate NetworkX minimum-cut encoding.
- 80 random request traces with tied arrivals: all 64 profiles per trace checked against event replay, unrestricted cut versus brute force, and a coherence-constrained cut versus the dynamic program.
- 20 additional traces: the event-age optimizer compared with exhaustive integer timeouts from 0 through 10 seconds, including values absent from its grids.
- 30 random five-node problems, each at 15 budgets: coherent frontier interpolation versus a linear program over all feasible policies (450 comparisons).
- Randomized replay accounting and explicit checks for zero timeout, simultaneous requests, terminal censoring, orphaned blocks and decimal boundaries.

For every full-trace fitted profile in the closure studies, an expiration-heap replay independently recomputes training hits and occupied time and checks the optimization result. This tests the accounting as well as the solver.

The implementation uses Python 3.12, NumPy and PyMaxflow [8]. Flow capacities are double precision; the proofs establish mathematical exactness, while the code provides numerical checks rather than an exact-arithmetic certificate. The largest absolute flow/objective residual among the 120 native fixed-grid cuts is 1.64×10^{-11} saved-block units. Returned cut labels are reduced to the smallest thresholds preserving their selected training hits, removing arbitrary zero-weight retention choices. Machine-specific timings are single observations on an ARM64 macOS system and exclude data download; they are not a hardware benchmark.

7 Results

7.1 The coherent restriction is nearly exact on these traces

Across three native traces, four groupings and ten prices, the unrestricted fixed-grid optimum equals the coherent optimum to numerical tolerance in 118 of 120 cases. The two exceptions are the tool-agent group and group-depth models at $\lambda = 0.01$, each improving training utility by 4.718 saved-block units, or 3.09×10^{-5} per training input block. The largest corresponding test utility improvement over the same-grouping coherent policy is 2.79×10^{-5} per input block. Thus the exact solver predominantly certifies a simpler policy on these workloads; it does not demonstrate a large empirical advantage from abandoning coherence.

The independent-local-hit upper bound can be less informative. For synthetic group-depth retention at $\lambda = 0.003$, the exact optimum is 599.160 while the raw-hit bound is 865.498. At $\lambda = 0.03$, the exact optimum is zero while the raw-hit bound remains 31.636. The bound is valid but can ascribe value to incompatible local hits. Reporting it together with the attainable coherent solution prevents that value from being mistaken for reusable-prefix savings.

7.2 Heterogeneity helps in specific storage-price regimes

Figure 2 reports every selected-policy contrast; the complete utility table is in Appendix A. Tool-agent utility is greater than the global-TTL baseline at all nine positive prices. Conversation utility is lower at 10^{-4} and 3×10^{-4} , higher at the remaining seven positive prices, and tied at zero. Synthetic utility is lower at prices from zero through 10^{-3} , higher at 0.003, and tied at the five largest prices. This variation rules out a blanket claim that more detailed profiles transfer better.

As an interpretable operating point rather than a selected headline, at $\lambda = 0.01$ on tool-agent test data the selected policy reuses 37.113% of full input blocks with 69.85 mean resident blocks. Global TTL reuses 37.023% with 715.97 mean blocks. The 90.2% reduction in mean occupancy accompanies a similar hit fraction in this replay. However, group TTL already achieves 37.139% with 69.83 mean blocks and has slightly greater utility than the selected policy. Most of this operating point’s improvement therefore comes from prefix grouping, an established idea, rather than the additional depth structure.

At larger tool-agent prices, depth structure becomes more useful: the selected policy exceeds group-TTL utility by 0.00521, 0.03023 and 0.06094 per input block at prices 0.1, 0.3 and 1, respectively. The synthetic trace gives the counterpoint: at zero price the selected policy’s hit fraction is 0.68154, compared with 0.70793 for global TTL. The training-optimal short profiles do not preserve all reuse patterns appearing later. Validation selection between the two structured classes does not protect against a global baseline outperforming both.

Figure 3 and Appendix B provide the capacity-policy context. For example, tool-agent LRU at capacity 256 attains 0.36355 hit fraction and 255.38 mean blocks. At capacity 16,384 it attains 0.53841, whereas the no-eviction reference is 0.57223. These comparisons describe the observed tradeoffs and do not establish dominance at matched hard capacity.

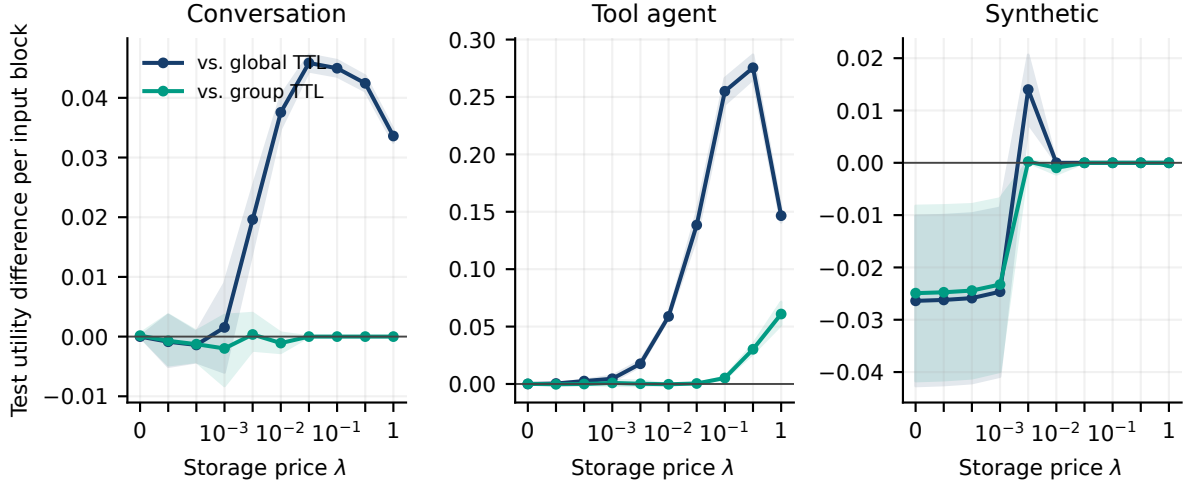


Figure 2: Held-out utility difference of the validation-selected coherent policy against global and group TTL, at all ten declared prices. Shaded regions are descriptive paired 60-second-bucket bootstrap intervals, not simultaneous bands. The horizontal positions enumerate the declared prices; the axis is not linear in price. Negative differences are retained.

Trace	Grid nodes	Event nodes	Grid solve (ms)	Event solve (ms)	Prep. (s)
Conversation	4,694	28,280	4.25	24.41	0.62
Tool agent	4,341	26,901	4.00	23.44	0.68
Synthetic	2,362	9,995	1.68	8.47	0.20

Table 2: Group-depth graph size and median solve time over ten prices. Preparation time is for the event-age graph and cost tables. Solver calls use already prepared inputs; timings are single-run descriptive measurements.

7.3 Continuous optimality and transfer are different properties

The event-age construction optimizes all nonnegative static group-depth timeouts on training. Figure 4 shows its full comparison with fixed-grid PCR. Training utility never decreases, as required by Theorem 3. Tool-agent test improvements reach 0.02264 utility per input block at price 0.3, but the event-age profile is worse at prices zero and 0.003. On the synthetic trace, its test utility is 0.00925 lower at price 0.003 even though its training utility improves. Conversation transfer is also mostly weaker. Exact empirical optimization is therefore an oracle for a specified historical objective, not a guarantee against workload shift or overfitting.

7.4 Computation, budgets and timestamp sensitivity

Table 2 gives graph sizes and observed solver times. Compression reduces fixed-grid group-depth graphs to 2,362–4,694 nodes, while full event-age graphs use 9,995–28,280 nodes. These sizes make the method suitable as an offline configuration oracle for the released traces. The coherent frontier calculation finds 59, 66 and 11 supported group-depth profiles on conversation, tool-agent and synthetic training data, respectively. Their certified mixture frontiers appear in Figure 5. These are training guarantees only.

Coalescing arrival times to whole seconds and refitting the same fixed-grid policies changes test utility by at most 0.000404, 0.009114 and 0.001198 across all groupings, modes and prices for conversation, tool-agent and synthetic data, respectively. This checks sensitivity to timestamp resolution and batching. It does not simulate positive request duration: a block still becomes available after its batch in the coalesced replay. The maximum tool-agent change occurs outside

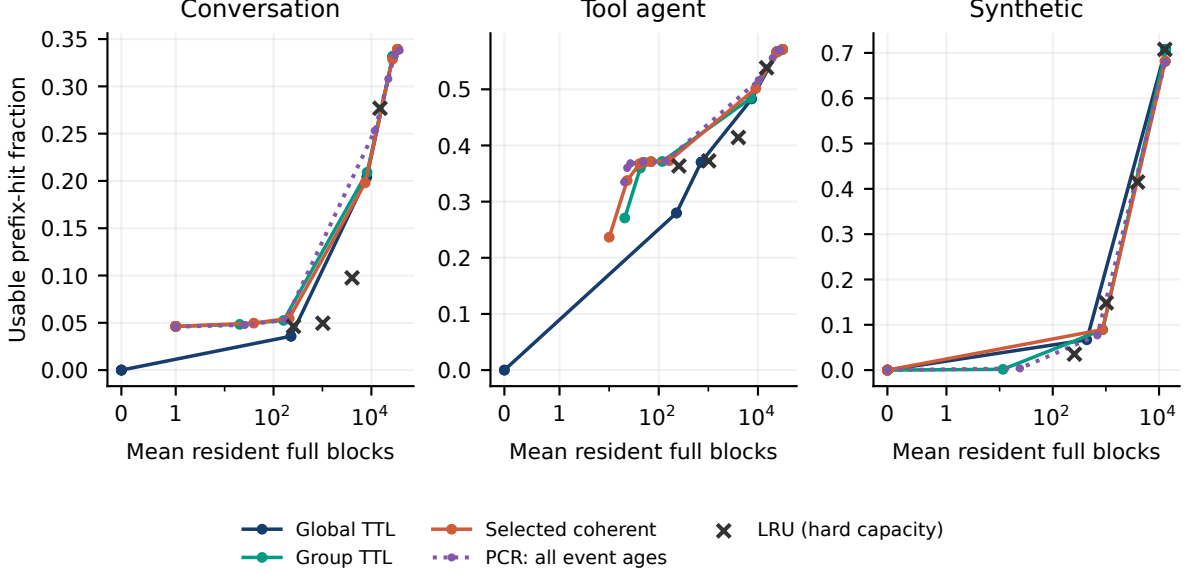


Figure 3: Realized test hit fraction versus mean occupancy. TTL points are fitted on training at the declared prices; connecting them is a visual guide, not a certified test frontier. PCR with all event ages is exploratory. Crosses show hard-capacity LRU at four capacities, plotted at its realized mean occupancy; TTL and LRU do not have equivalent instantaneous memory guarantees. The synthetic workload exposes weak transfer of structured static profiles.

the group-depth PCR curve, whose maximum absolute change is 0.001449.

8 Industrial use and limitations

A practical use of PCR is a retention-planning and audit component for an independently addressed KV store. Given recent access logs, it can return a timeout profile, its fitted reward and occupied time, the coherent lower bound, and the local-hit upper bound. If the bounds agree, an operator has a concrete reason to deploy the simpler coherent profile. If they differ, the exact graph quantifies the historical opportunity before any engine integration. Repeated offline fitting and a shadow replay on subsequent windows are plausible next steps, but their value is not established by the present one-window evaluation.

The unit reward counts saved full prefix blocks. Actual prefill cost depends on position, model architecture, batching, kernels and transfer overhead. The graph accepts exogenous nonnegative per-hit rewards, but a reward that changes with the chosen profile or queue state requires a different model. Similarly, a store that automatically deletes descendants when an ancestor expires does not follow our unrestricted residency model. The coherent variant is compatible with the prefix-residency invariant, subject to the engine’s other constraints.

The experiments use short, sampled workloads from one public release, one chronological split, and a deliberately small family of transferable groups. There is no claim of optimal online adaptation, generalization to other model families, or measured dollar savings. Peak occupancy is reported because an attractive mean can coexist with an infeasible instantaneous requirement. The graph objective contains no hard-capacity constraint. Its mathematical certificate also assumes that accesses refresh every referenced full block; selective writes would require revised accounting.

The next systems experiment should record block completion and transfer times, active-request pinning, actual peak allocations and model-specific prefill cost, then compare policies

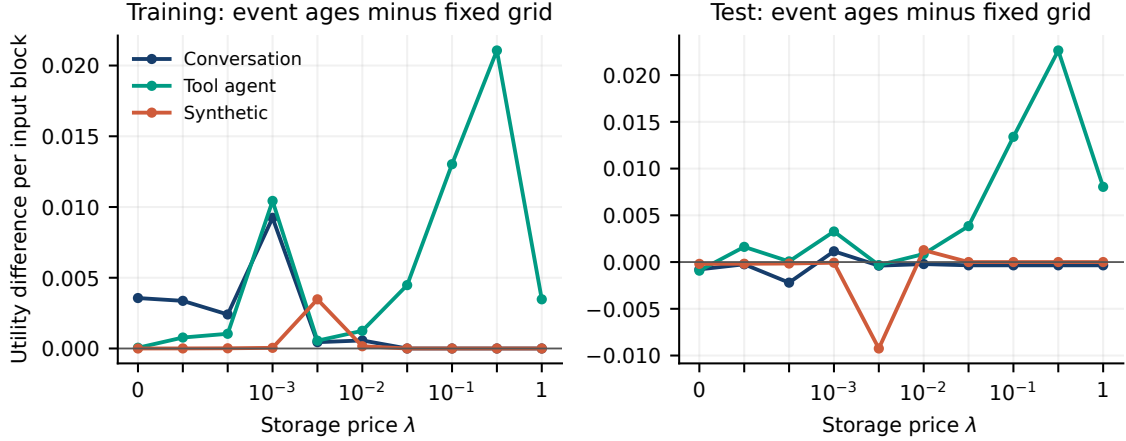


Figure 4: Exploratory event-age PCR versus the frozen 13-choice grid, using the same group-depth class. Training gains include the effect of allowing timeouts above 900 seconds. Exact fitting does not uniformly improve the later window.

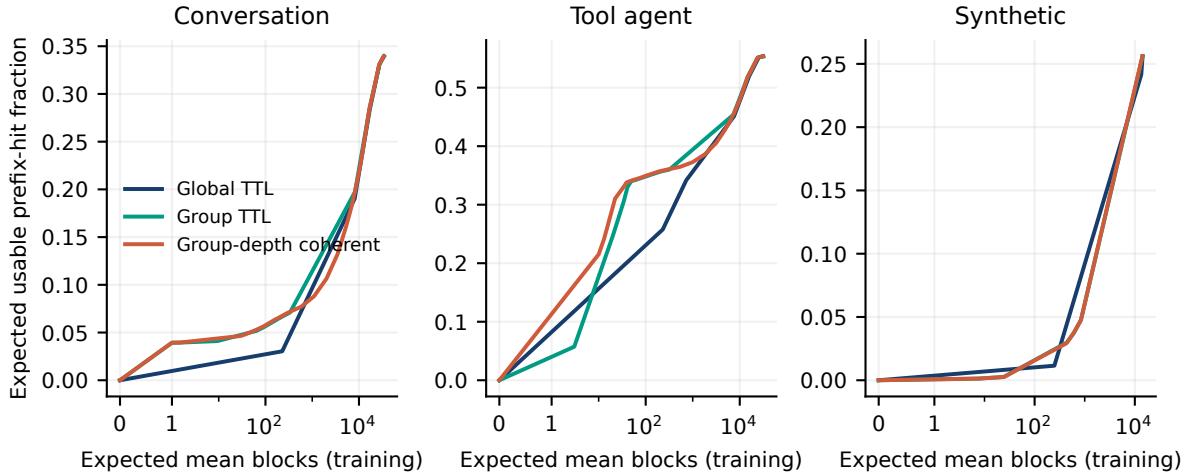


Figure 5: Certified training frontiers for mixtures of coherent policies. Lines interpolate whole-profile randomizations under an expected memory-time budget. They are not guarantees on peak residency, nor held-out performance.

at a common capacity and service-level objective. The present code establishes a checked optimization component for that experiment.

9 Conclusion

Usable prefix reuse is a conjunction of local retention events. Making those prerequisites explicit turns a seemingly coupled static-timeout problem into a maximum-weight closure, and observed reuse ages suffice for continuous timeout optimization. The resulting oracle can both optimize unrestricted retention and certify a simpler coherent policy. Public-trace results show substantial room for differentiated retention in some storage-price regimes, little additional benefit from dropping coherence on the fixed grid, and nonuniform transfer from finer empirical fitting. These distinctions make the method useful as a precise configuration model and benchmark without conflating mathematical optimality with unmeasured system performance.

Data and software availability

The repository contains the optimizer, independent replay, exhaustive tests, pinned trace-download manifest, all fitted profiles and result tables, figure generation, and manuscript source. Raw traces are obtained from their original Mooncake release rather than redistributed. The initial local protocol, exploratory extensions and implementation corrections are recorded separately.

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A Complete declared-price utility table

Table values are $(H - \lambda C)/M$ on the test window. “Selected” denotes validation-selected coherent depth or group-depth retention. “PCR grid” is unrestricted group-depth retention on the frozen grid; “PCR ages” uses every training reuse age. Rows are generated directly from the result files.

Trace	λ	Global	Group	Selected	PCR grid	PCR ages	Sel. – Global
Conversation	0	0.33917	0.33901	0.33917	0.33875	0.33796	0.00000
Conversation	0.0001	0.29703	0.29688	0.29617	0.29578	0.29554	-0.00086
Conversation	0.0003	0.22758	0.22746	0.22619	0.22596	0.22376	-0.00140
Conversation	0.001	0.10211	0.10563	0.10365	0.10365	0.10480	0.00154
Conversation	0.003	0.02721	0.04647	0.04682	0.04682	0.04644	0.01961
Conversation	0.01	0.00711	0.04577	0.04468	0.04468	0.04447	0.03757
Conversation	0.03	0.00000	0.04584	0.04584	0.04584	0.04549	0.04584
Conversation	0.1	0.00000	0.04496	0.04496	0.04496	0.04461	0.04496
Conversation	0.3	0.00000	0.04243	0.04243	0.04243	0.04209	0.04243
Conversation	1	0.00000	0.03360	0.03360	0.03360	0.03326	0.03360
Tool agent	0	0.57121	0.57110	0.57121	0.57091	0.56999	0.00000
Tool agent	0.0001	0.54504	0.54573	0.54544	0.54545	0.54707	0.00040
Tool agent	0.0003	0.49983	0.50238	0.50236	0.50236	0.50243	0.00253
Tool agent	0.001	0.41582	0.41946	0.42037	0.42037	0.42364	0.00455
Tool agent	0.003	0.35097	0.36837	0.36859	0.36859	0.36826	0.01762
Tool agent	0.01	0.30602	0.36513	0.36486	0.36489	0.36575	0.05884
Tool agent	0.03	0.21846	0.35645	0.35688	0.35688	0.36073	0.13843
Tool agent	0.1	0.07546	0.32522	0.33043	0.33043	0.34383	0.25497
Tool agent	0.3	0.00000	0.24521	0.27544	0.27544	0.29808	0.27544
Tool agent	1	0.00000	0.08566	0.14660	0.14660	0.15465	0.14660
Synthetic	0	0.70793	0.70646	0.68154	0.68007	0.67988	-0.02639
Synthetic	0.0001	0.69913	0.69767	0.67292	0.67146	0.67128	-0.02621
Synthetic	0.0003	0.68154	0.68010	0.65567	0.65423	0.65407	-0.02587
Synthetic	0.001	0.61995	0.61858	0.59531	0.59393	0.59386	-0.02464
Synthetic	0.003	0.05826	0.07203	0.07226	0.07226	0.06301	0.01401
Synthetic	0.01	0.00000	0.00097	0.00000	0.00097	0.00224	0.00000
Synthetic	0.03	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Synthetic	0.1	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Synthetic	0.3	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000
Synthetic	1	0.00000	0.00000	0.00000	0.00000	0.00000	0.00000

B Capacity baselines and unbounded reference

Capacity is measured in full blocks. All rows use a cold test-window start. FIFO and LRU are capacity-limited hash-store baselines. Their mean occupancy is not a separate capacity guarantee for the TTL methods.

Trace	Policy	Capacity	Hit fraction	Mean blocks	Peak blocks
Conversation	LRU	256	0.04625	255.33	256
Conversation	FIFO	256	0.04259	255.33	256
Conversation	LRU	1024	0.04957	1018.52	1024
Conversation	FIFO	1024	0.04720	1018.52	1024
Conversation	LRU	4096	0.09753	4026.41	4096
Conversation	FIFO	4096	0.09245	4026.41	4096
Conversation	LRU	16384	0.27686	15003.35	16384
Conversation	FIFO	16384	0.24975	15003.35	16384
Conversation	Unbounded	–	0.34339	38862.20	73542
Tool agent	LRU	256	0.36355	255.38	256
Tool agent	FIFO	256	0.33270	255.38	256
Tool agent	LRU	1024	0.37253	1018.67	1024
Tool agent	FIFO	1024	0.36279	1018.67	1024
Tool agent	LRU	4096	0.41423	4024.92	4096
Tool agent	FIFO	4096	0.40727	4024.92	4096
Tool agent	LRU	16384	0.53841	14909.48	16384
Tool agent	FIFO	16384	0.51924	14909.48	16384
Tool agent	Unbounded	–	0.57223	35768.20	67335
Synthetic	LRU	256	0.03526	255.69	256
Synthetic	FIFO	256	0.03662	255.69	256
Synthetic	LRU	1024	0.14841	1016.80	1024
Synthetic	FIFO	1024	0.15305	1016.80	1024
Synthetic	LRU	4096	0.41506	3942.27	4096
Synthetic	FIFO	4096	0.39546	3942.27	4096
Synthetic	LRU	16384	0.70769	12824.97	16384
Synthetic	FIFO	16384	0.69476	12824.97	16384
Synthetic	Unbounded	–	0.70793	13317.67	18074

C Reproduction and audit details

The repository’s **Makefile** separates data download, tests, experiments, figures and typesetting. The source revision and file digests are recorded in **data/manifest.json**; experiment outputs include complete price sweeps, labels, raw and usable hit counts, mean and peak occupancy, and runtime metadata. The prospective protocol digest is

3f47a960f95a3f7b13fbf68dfc8fab51fde3f12a2f9b7422b0a06497b9a31719.

This digest identifies the local plan, not independent registration. Timing columns can vary between runs; the replay endpoints and fitted profiles are deterministic for the declared inputs and numerical environment. Test and validation caches are intentionally cold rather than initialized from training. No hidden warm-start state is included in the reported savings.

Integer time comparisons avoid disagreements between subtraction-based age tests and addition-based expiration tests at decimal boundaries. Source times have millisecond precision; calculations use integer microseconds so fractional split boundaries remain representable. Event-grid rounding is downward to actual training ages, which explains why tied training optima can transfer differently from longer fixed-grid timeouts. All numerical results in this manuscript were regenerated after these implementation corrections.